Procedural Ocean: Design of Maritime Visualizations using Game Technology

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***Abstract*—This paper presents Procedural Ocean, a technical platform that integrates procedural content generation with game engine technology to transform geospatial datasets into interactive 3D marine visualizations while maintaining both data fidelity and real-time performance. We present a three-layer architecture which includes a core innovation, our Procedural Rule Pack (PRP) system enables modular generation of complex interdependent maritime behaviors from real-world data sources.**

Through a research-through-design methodology, we document critical technical trade-offs between scientific accuracy, computational performance, and stakeholder accessibility. These contributions provide transferable architectural patterns for developing data-driven serious games requiring real-time procedural content generation.

***Index Terms*—Procedural Content Generation, Maritime Environments, Serious Games**

I. Introduction

Maritime Spatial Planning (MSP) is a comprehensive approach to managing the increasingly complex use of ocean spaces, aiming to balance environmental protection with economic development in sectors such as renewable and non-renewable energy, shipping, fishing and aquaculture [12]. The European Union's 2014 directive mandating MSP by 2021 has accelerated the need for advanced tools that can address the complexities of maritime governance [15]. These challenges stem from the dynamic nature of maritime environments, intricate interactions between human activities and ecosystems, and the need to coordinate diverse stakeholders with often conflicting interests across international boundaries [8]. Still, the more traditional planning approaches struggle to address the scale and interconnectedness of maritime systems, where decisions in one sector inevitably affect others, and local interventions may have basin-wide consequences [19].

Recently, serious games have emerged as promising solutions for engaging stakeholders and facilitating collaborative decision-making in MSP processes [25]. By transforming complex maritime planning scenarios into interactive gaming experiences, these systems enable multi-stakeholder participation, strategic thinking, and experiential learning about ocean environments [35]. However, most current MSP tools, including serious games platforms, still rely on two-dimensional Geographic Information Systems (GIS) interfaces, which lack experiential realism and fail to capture the three-dimensional complexity of maritime environments [4], [17]. This type of representation creates psychological distance between stakeholders and the ecosystems they manage, making it difficult to fully appreciate the scale of proposed developments, assess local and cumulative impacts, or understand inter-dependencies among maritime sectors. Consequently, critical interconnections are overlooked, resulting in unintended consequences, inefficient resource management, or intensified stakeholder conflicts.

This research addresses the question: *How can procedural generation efficiently transform geospatial data into performant, accurate 3D interactive environments?* To answer this, we present Procedural Ocean, a proof-of-concept platform that integrates real geospatial data with procedural content generation. Procedural Ocean aims to complement existing MSP tools by offering a three-dimensional, interactive, and immersive environment that bridges the gap between data-driven analysis and experiential understanding. The platform supports scenario assessment, allowing users to visualize how planning decisions affect maritime environments and to explore interdependencies between natural systems and human activities [28].

Rather than presenting Procedural Ocean solely as a novel system, this paper's primary contribution is a detailed examination of the design decisions, trade-offs, and architectural patterns necessary to adapt game engine technologies for scientific maritime visualization. We provide an analysis of how procedural generation techniques, traditionally used for entertainment, can be rigorously applied to maintain scientific accuracy while achieving real-time performance. This 'research-through-design' approach [23] offers transferable insights for researchers developing similar immersive scientific visualization platforms.

Our methodology focuses on developing and evaluating a proof-of-concept application that can simulate large-scale maritime environments in a visually realistic and computationally efficient manner. This paper presents the definition of the requirements and the architecture prioritizes interoperability, optimization, and extensibility, supporting standard geospatial data formats such as GeoJSON and GeoTIFF, which are widely used in maritime planning workflows [7]. While the current implementation focuses on the North Sea, the modular design enables easy adaptation to other regions worldwide. We critically examine the trade-offs involved in procedurally generating detailed maritime environments from real-world data [3], [36].

Our main contributions include a flexible architecture for integrating heterogeneous spatial data, multi-scale procedural generation techniques for above and underwater environments, and data-driven solution for simulating interdependent human activities. We also discuss current limitations, laying the groundwork for future advancements in using game technologies to develop better serious and applied games.

II. Literature Review

A. Serious Games

Serious games have demonstrated significant potential for stakeholder engagement and collaborative decision-making in environmental planning contexts. Platforms like MSP Challenge have successfully facilitated multi-stakeholder workshops across European sea basins by transforming complex maritime planning into accessible, interactive gaming experiences [27], [35]. These systems excel at creating shared mental models among diverse participants and fostering collaborative problem-solving through game mechanics that encourage strategic thinking and perspective-taking [25], [43]. However, current serious games for environmental applications face significant technical limitations in accurately representing real-world environments. Most existing platforms rely on abstract symbols, simplified 2D maps, or basic 3D representations that fail to capture the spatial complexity and scale of natural systems [20]. This representational gap creates what Klain and Chan [26] describe as "cognitive disconnect," where three-dimensional ecosystems are reduced to abstract game elements, limiting players' ability to develop accurate spatial mental models. Furthermore, the technical challenge of integrating real-world scientific datasets with game engines while maintaining interactive performance remains largely unaddressed [13].

B. Digital Twins

The concept of digital twins (DTs) has emerged as a transformative framework for understanding and managing marine ecosystems. Ocean digital twins constitute consistent, high-resolution, multi-dimensional, near real-time virtual representations of marine environments, integrating observations, artificial intelligence, and advanced modeling on high-performance computing infrastructure. The European Digital Twin of the Ocean project [1] exemplifies this approach, creating comprehensive digital replicas of global maritime environments encompassing both natural and anthropogenic components [10]. While these systems excel at data integration and predictive modeling, their visualization interfaces typically rely on traditional web-based dashboards and 2D mapping systems that fail to convey the true scale and complexity of three-dimensional marine environments [40]. This limitation constrains the potential of digital twins to provide intuitive, immersive experiences that could enhance understanding of complex maritime systems and their interdependencies.

[1] www.digitaltwinocean.mercator-ocean.eu, 2025

C. Procedural Content Generation Using Real-World Data

Procedural content generation (PCG) has become vital for the creation of large-scale, data-driven virtual environments, and provide scalable methods for simulating complex geographies and phenomena, by integrating real-world data. Existing research demonstrates how remote sensing data can constrain and inform procedural algorithms [11], enabling the generation of virtual terrains that maintain physical plausibility. Instead of direct replication, real-world data are used as constraints and input parameters.

D. Virtual World Simulations

Recent advances in virtual world technology have enabled impressive maritime visualizations, yet challenges remain in creating adaptable systems for diverse underwater environments. Projects like OceanViz [39] and NOAA's Virtual Ecosystem Viewer [29] demonstrate progress in translating complex ecosystem data into 3D underwater experiences. These tools effectively bridge the gap between scientific models and stakeholder understanding by visualizing marine populations and environmental changes. However, both systems are primarily a visualization tool constrained by their dependence pre-computed model scenarios.

III. Design Requirements

Substantial progress has been made in serious games, digital twins and procedural generation, and the complementary nature of these technologies. The Procedural Ocean platform attempts to take advantage of the best qualities of all, and is designed not to replace, but to complement existing platforms. Before the technical design, it is essential to identify and define the target audience alongside the primary application goals and requirements. These elements serve as guiding principles throughout the development process. In our design process we defined our stakeholder and target audience as: maritime spatial planners; government officials and policymakers; industry representatives, conservation sectors, and educational institutions.

A. Requirements

Our design process was guided by principles that were derived from stakeholder needs and requirements [1], prioritizing both functional capabilities and user experience. The following key requirements were established:

- **Realism**: Visual replication of real-world maritime conditions including dynamic ocean and atmospheric modeling, realistic light propagation, and accurate underwater physics, as well as anthropogenic and biome elements.
- **Performance**: Efficient simulation and displaying of complex ocean dynamics, including fluid and atmospheric conditions, for a comfortable user experience.
- **Standard Sources**: Integration with existing GIS data sources and APIs, accommodating heterogeneous maritime datasets with varying quality and formats. Mechanisms for data validation and traceability to address data quality concerns.

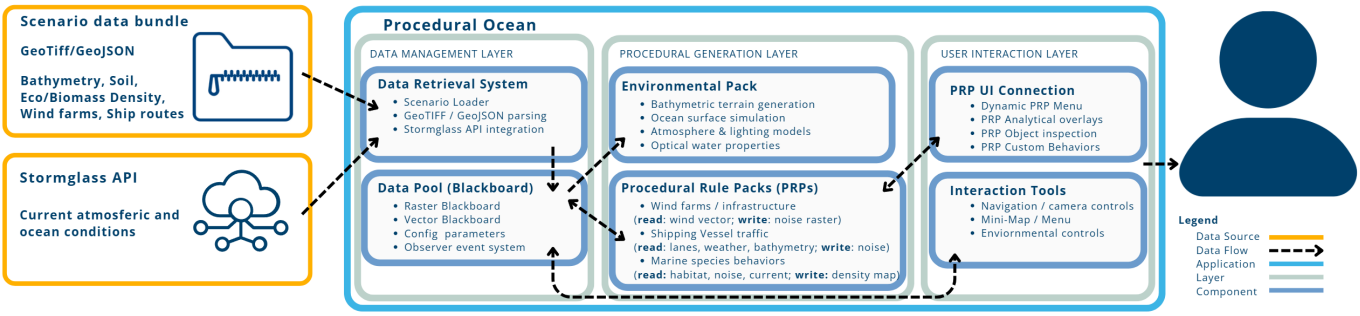


Fig. 1. Procedural Ocean architectural 3-layer design and component organization including data sources and data flows.

- **Extensibility**: Easily extendable to support local dynamics, regional customizations, and future model additions to facilitate modeling of marine ecosystems.
- **Interdependency**: Allow the definition and simulation of diverse behaviors, such as predator-prey dynamics, vessel traffic, and interactions with environmental changes.
- **Open Integration**: Integration with other marine tools and workflows, allowing it to serve as an external visualization or decision-support system.

IV. Development Details

Following the design science research paradigm [22], we adopted a build-and-evaluate cycle to develop a proof of concept platform based requirements that addresses the identified gap, and can confirm the feasibility and identification of the limitations of such applications. This methodology aligns with established frameworks for developing interactive systems where technical feasibility must be balanced with user accessibility [30], [32]. In this section we detail the design and implementation details and justify their choices.

A. Development Tools

The implementation of Procedural Ocean required development tools capable of real-time 3D visualization, accuracy, and cross-platform deployment. After evaluating alternatives, we selected Unreal Engine 5 as the core platform, given its modular architecture with C++ API access for custom maritime simulation components, and physically-based rendering with custom shaders enabling realistic light attenuation and surface dynamics essential for authenticity [14], [31]. Its active community and asset store further accelerated development.

To support agile methodology and stakeholder testing, we established a CI/CD pipeline. Automated tests for data import and performance validation, combined with nightly builds deployed to a secure portal, enabled non-technical users to continuously validate features without local setup, ensuring rigor while allowing rapid iteration [38].

B. System Architecture

The Procedural Ocean platform employs a three-layer architecture designed to separate concerns and ensure modularity, scalability, and maintainability [6]. The Procedural Ocean architecture is depicted in Fig. 1 and shows: *i)* an independent Data Management Layer responsible for data acquisition and storage, *ii)* a Procedural Generation Layer that processes data, generates and updates 3D content, and *iii)* a User Interaction Layer managing visualization modes and user interfaces. This separation enables independent development, responsibility, to manage the different 3D scenarios with different requirements.

1) Data Management Layer: The data management layer comprises two primary components: a data retrieval system and a data pool storage structure. The goal is to create a versatile data management layer, that is able to load data from heterogeneous sources, including GIS data and other external APIs, and make it easily accessible to the other components with support change propagation.

a) Data Retrieval System: The data retrieval system is responsible for loading and parsing a scenario data bundle. Scenarios are defined through a compressed archive that contains:

- A manifest file with scenario metadata including name, geographic bounds, and references to local or external geo-located data files.
- Local Raster data in GeoTIFF format [33] for continuous variables such as bathymetry, environmental conditions, and species density distributions.
- Local Vector data in GeoJSON format [7] representing discrete features including wind farm boundaries, submarine cables, shipping lanes, and maritime protected areas.

Scenario data bundles contain all long-term data locally using a folder structure, to support organization and long-term archiving. In addition, the Data Retrieval implements a hierarchical fallback mechanism for data availability. When specific data is unavailable in the scenario, the system queries an internal Procedural Ocean repository based on geographic coordinates, this repository contains high detailed data from some regions to complement incomplete scenario bundles. If essential data is unavailable or coordinates fall outside defined bounds, default values are used. For example, seafloor sediment maps is frequently not available, as such, it would be initialized with the most commonly found seafloor, namely sand. Any data that are not part of the data bundle are appropriately logged and users notified.

A secondary information source complements the scenario data bundle. The system retrieves real-time environmental



Fig. 2. Procedural Ocean can obtain the environment conditions using Stormglass API, or manually adjust time-of-day, wind speed and direction, cloud coverage and precipitation.

conditions through a RESTful API of Stormglass API [41] to know current meteorological and oceanographic parameters including wind, current, wave, temperatures, and atmospheric conditions. This real-time integration ensures that simulations reflect actual environmental conditions enhancing relevance for operational scenarios.

b) Data Pool: The parsed data is organized within a data pool component consisting of a dictionary structure containing named blackboards [21], each responsible for specific data domain. Blackboards provide an effective mechanism for managing shared data in systems with multiple independent processing components. Individual blackboards store either atomic values (e.g., simulation time, global environmental parameters) or complex data structures. Raster blackboards maintain gridded spatial data utilizing efficient spatial indexing for rapid queries [34]. Vector blackboards store geometric features with associated attributes, supporting topological operations required for spatial analysis. Other models were considered, such as the event bus or the publisher/subscriber pattern, and although these approaches offer greater data encapsulation, they are less memory efficient and computationally performative.

Each blackboard incorporates a comprehensive metadata system using multi-keyword tagging to facilitate discovery and categorization. This semantic organization is for both automated processing and user interface grouping [37].

The data pool also implements an observer pattern [16] through a registration system that propagates change notifications via callback functions. This event-driven architecture eliminates the need for continuous polling or redundant regeneration, crucial to maintaining real-time performance requirements.

2) Procedural Generation Layer: The procedural generation layer consists of two complementary components: the Environmental Pack responsible for generation of the static base environment, and the Procedural Rule Pack system managing dynamic content generation on top of the environmental pack.

a) Environmental Pack: The Environmental Pack generates foundational environmental elements following a hierarchical approach. It starts with generation of the bathymetric seabed creating detailed geometry from raster elevation data using an iterative process to create an up-sampled and normalized 3D mesh with accurate UV maps. The component also generates a corresponding texture map using sediment type data information to enhance visual realism while maintaining data accuracy.

Following terrain generation, the environmental pack parameterizes the other environmental conditions including time of day, atmospheric conditions (cloud coverage, precipitation, visibility), ocean state (wave and current patterns), and underwater optical properties (light attenuation, turbidity). These parameters drive physically-based simulations for realistic ocean and atmospheric rendering (see Fig. 2 for reference).

b) Procedural Rule Packs: Procedural Rule Packs (PRPs) function as modular plugins, implementing standardized interfaces to ensure compatibility while enabling specialized content generation. Each PRP encapsulates procedural rules for generating and managing specific asset categories, from maritime life to human infrastructure.

In addition, each PRP can implement multiple visualization modes for the same underlying data, supporting different analytical needs. For example, fish population data can be visualized through animated 3D schools for immersive exploration or density heatmaps for quantitative analysis. This multi-modal approach addresses the diverse needs of stakeholders from different backgrounds. Fig. 3 illustrates some existing PRPs, and Fig. 4 shows a variety of display modes.

Important to note that PRPs not only use and combine the data from the blackboards, but also create and modify blackboards. This way a PRP is technically isolated from the others while be able to still propagate dependencies to through the data pool. For instance, wind speed affects both the energy output of wind turbines and the noise they generate. This noise is reported by the wind farm PRP in the data pool and can impact other PRPs responsible for managing marine life. In this way, deep inter-dependencies can form. However, circular dependencies are prevented by the data pool, while still allowing for complex, emergent behaviors.

3) User Interaction Layer: The user interaction layer implements a unified interface supporting both desktop and immersive modalities. The design prioritizes intuitive interaction while providing access to analytical tools through progressive disclosure. Navigation uses industry-standard controls with time manipulation capabilities for simulation pacing. The menu system employs layered complexity, enabling non-technical stakeholders to explore intuitively while provid-

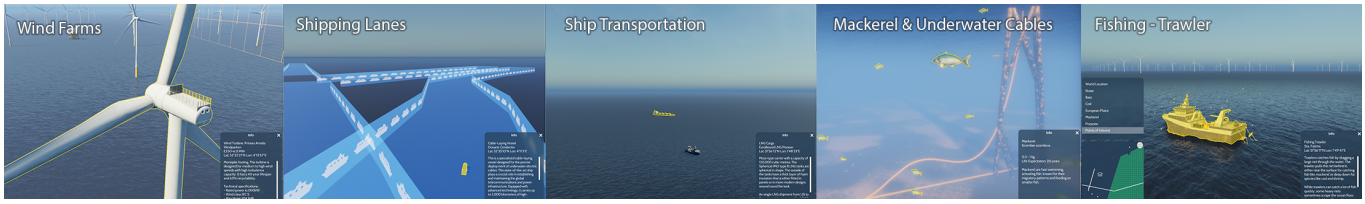


Fig. 3. Example of existing PRPs that cover some human activities (wind turbines, transformer station, cables, fishing, shipping), biome (porpoise, cod, makerel) and visualization modes (shipping lanes).

ing advanced users with comprehensive analytical tools and weather controls (Fig. 2).

a) Asset Interaction: PRP assets respond to selection with contextual information displays, such as vessel specifications or wind turbine energy output. Selected assets can trigger detailed demonstrations, like wind turbine assembly sequences or fishing gear operations, providing experiential understanding of maritime activities and their spatial relationships within the broader ecosystem context. The modular PRP architecture enables extension and customization for future applications, allowing adaptation to specific serious game objectives and target audiences from educational scenarios.

C. Integration Capabilities

To extend the value of Procedural Ocean beyond standalone operation, we consider the integration with existing maritime spatial planning tools.

a) Command-Line Interface: To allow this integration, we developed a command-line interface (CLI) that enables programmatic invocation of Procedural Ocean from external applications. The CLI accepts standard parameters including scenario configuration files, geographic bounds, visualization preferences, and output specifications. This design follows established patterns for scientific software tools integration [42]. While the CLI successfully enables workflow integration, its adoption may require modification of existing tools and technical expertise.

b) Web-Based Scenario Generation: To facilitate rapid scenario creation, we also implemented an online web service coupled with online geoserver infrastructure. This system automates scenario bundle generation by aggregating data from multiple sources based on user-specified parameters, including geographic bounds, and required data layers. The service automatically queries relevant data sources, including external APIs with OGC-compliant Web Map Services (WMS) and Web Feature Services (WFS), ensuring compatibility with standard geospatial data infrastructures [9], performs necessary transformations, and packages results into the compressed scenario bundle expected by Procedural Ocean. This approach significantly reduces the technical barrier for scenario development while ensuring access to current data from authoritative sources, and can provide a base for experts to alter such scenarios.

c) MSP Challenge Platform Integration: A significant validation of Procedural Ocean's interoperability came through collaboration with the MSP Challenge development team [2], [35]. This partnership enabled seamless integration between the two platforms, leveraging the complementary strengths of both systems. MSP Challenge serves as a serious gaming platform for maritime spatial planning, enabling multi-stakeholder scenario exploration across various sea basins. Through our collaborative development effort, MSP Challenge users can select a subarea in the 2D map, which exports a data bundle containing their planning decisions—such as new offshore wind farms, marine protected areas, or fishing policy zones—and seamlessly transition to view and interact with these scenarios in Procedural Ocean's realistic 3D environment. This collaboration demonstrates how different serious gaming platforms can be integrated to enhance the overall stakeholder experience, combining strategic planning capabilities with immersive visualization.

V. Discussion

This section reexamines the identified requirements and evaluates our findings against the initial objectives. The primary motivation was to overcome the inherent limitations of 2D GIS-based maritime spatial planning tools. Our implementation demonstrates that is possible to develop a system using procedural generation to address the shortcomings identified in existing systems related to cognitive disconnect: psychological distance, scale misrepresentation, and inability to visualize inter-dependencies.

a) Visual Realism: The environmental pack is able to simulates dynamic ocean surfaces, e.g. light attenuation that follows Beer-Lambert law [18]. Similar real-time weather integration through the Stormglass API enables current atmospheric and oceanographic conditions to influence the visualization. Note that seasonal variations, such as biomass migrations, were not modeled, as the input data consists of historical yearly aggregates rather than time-resolved records.

b) Performance Optimization: Real-time performance is particularly crucial for real-time visualizations, and achieved through a multi-layered optimization strategy. Data optimizations and update cycles where directly integrated in the architectural design, front loading the effort and reducing the effort of PRPs updates to a minimum. 3D scene optimization has been approached through an automated and manual process. Automatic Levels of Detail (LOD) were generated by the chosen game engine, and combined with hand-made LODs for assets where automatic generation was unsuitable. These LODs are shown dependent on the player distance. Underwater scenes use particle systems and shader-driven animations to

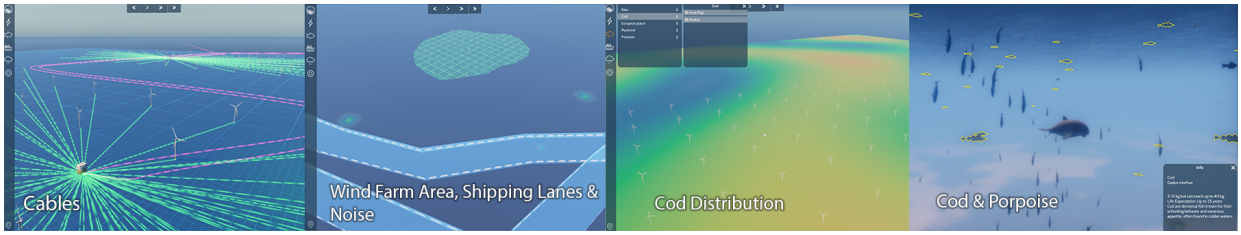


Fig. 4. Some examples of different visualization modes, that support more analytical work within Procedural Ocean.

visualize dynamic elements (benthic layer, fish schools, etc.) optimizing the rendering of large quantities of 3D assets. Further optimizations were achieved by the development team through dynamic asset generation based on the player position and view-direction, massively reducing the computational requirements. Finally, we optimized the materials and shaders through targeted engineering work, removing redundant nodes and streamlining shader graphs to achieve comparable visual results at significantly reduced computational cost.

Performance metrics demonstrate consistent frame rates above 60 FPS for typical planning scenarios (regions up to 100 km² with 2,500 active agents) on a laptop using a Core Ultra 9 185H Processor, 4080 laptop GPU, and 32GB LPDDR5 Ram. However, performance degrades with large agent densities (+7,000 agents), requiring dynamic agent aggregation techniques that trade individual accuracy for system responsiveness. Extended performance validation is ongoing and will be reported in future work.

c) Data Integration and Standardization: Procedural Ocean can be seamlessly integrated with existing MSP workflows, as it is able to intake standard geospatial formats (GeoJSON and GeoTIFF) from other established data pipelines. While NetCDF integration for temporal particle data was considered, we opted against simple data replay, as prerecorded particle movements would compromise the experience. Instead, we are currently developing optimized support for real-time particle simulation linked with the datapool, ensuring that processes such as water currents, algae blooms, or oil spill movements remain consistent.

Due to the variety of underwater habitats and significant variations in classifications, it is still required to create local/contextual PRPs. For instance, benthic habitat classifications vary substantially between European and North American systems. This finding aligns with broader challenges in maritime data standardization identified by the International Oceanographic Data and Information Exchange (IODE) community [24].

d) Extensibility: The modular architecture enables development of region-specific or activity-specific extensions without modifying core systems, allowing the platform to capture local dynamics. This design approach was successfully validated through multiple scenario implementations within the North Sea region. Future extensions are being considered for several applications: comparing different offshore energy production systems, supporting aquaculture planning, and expanding to the Baltic Sea with a brackish ecosystem with significantly lower salinity levels than typical marine environments.

e) Agent Behavioral Modeling: The requirement for interdependent agent behaviors was achieved through the data-driven blackboard system. Agents successfully respond to environmental stimuli—wind turbines adjust rotation based on wind data, marine mammals avoid noise sources, and fishing vessels modify routes based on weather conditions. The loose coupling between PRPs enables complex emergent behaviors without explicit inter-pack dependencies. However, behavioral model validation remains challenging. The current implementation relies on literature-derived parameters and expert knowledge, highlighting the need for standardized behavioral models.

f) External Integration Capabilities: The platform's potential as an external visualization system for existing MSP tools was partially achieved. The command-line interface enables scripted invocation with parameter passing, and the RESTful API supports remote scenario configuration. The MSP Challenge successfully integrated Procedural Ocean into their workflows for visualization of model outputs. However, full bidirectional integration with external tools is not supported, Procedural Ocean is a one-way data flow (external tools to Procedural Ocean) without callback mechanisms. Achieving true integration as a visualization service requires additional development of standardized MSP data exchange protocols.

A. Novel Contributions and Findings

a) Procedural Generation for Technical Visualization: Our implementation demonstrates that procedural content generation, traditionally associated with entertainment applications can be used to support maritime planning processes. The key innovation lies in using procedural techniques not for creating fictional content but for interpolating and extrapolating from sparse scientific observations.

The procedural architecture mirrors natural ecosystem dependencies: terrain and environmental conditions underpin human activities and biological distributions and enabling efficient computation and straightforward extensibility. Particularly successful was the multi-modal visualization capability within Procedural Rule Packs, which allows seamless transitions between photorealistic and analytical representations of the same data. This addresses diverse stakeholder needs

within a single platform, bridging disciplinary boundaries by letting scientists examine quantitative distributions while policymakers experience qualitative impacts.

b) Behavioral Modeling and Emergent Phenomena: The agent-based behavioral modeling system revealed unexpected insights into stakeholder understanding of marine ecosystems. The visual representation of behavioral responses—such as marine mammals avoiding noise from shipping lanes or wind farms—made abstract environmental impact concepts immediately comprehensible. This finding supports theories of embodied cognition in environmental understanding, particularly Barsalou's grounded cognition framework [5], which argues that spatial reasoning draws on sensorimotor experience rather than abstract symbols.

The loose coupling between Procedural Rule Packs through the Data Pool enabled emergent behaviors that enhanced system realism without explicit programming. For example, the interaction between shipping traffic patterns and fish school movements, mediated through noise propagation data, produced avoidance behaviors that stakeholders recognized as realistic without direct dependency between them.

B. Limitations and Challenges

Despite successful achievement of core objectives, several limitations merit discussion. First, the platform's effectiveness depends heavily on data availability and quality. While the fallback mechanism ensures operational stability, the use of default values in data-sparse regions may mask uncertainty in ways that could affect planning decisions. Future versions should incorporate explicit uncertainty visualization techniques, such as confidence interval overlays, opacity-encoded heat maps or soft boundary rendering for probabilistic distribution zones.

Second, validation of the platform's effectiveness as a serious games tool remains an ongoing effort. While initial stakeholder feedback indicates that PO successfully achieves its technical design objectives, comprehensive user validation studies are currently underway. These systematic user studies will be reported in future publications as they fall beyond the scope of this technical contribution.

Third, validation of behavioral models presents distinct challenges. Quantitative validation against observed ecosystem responses requires long-term monitoring data that is rarely available at the appropriate spatial and temporal scales needed for meaningful comparison. This challenge is not unique to our platform but reflects broader limitations in maritime observation infrastructure, highlighting the need for continued investment in comprehensive ocean monitoring systems to support both research and operational applications.

C. Future Directions

The successful implementation of Procedural Ocean demonstrates that it is possible to create procedural visualization technologies that transcend their origins in entertainment to provide valuable tools for environmental and maritime management. The platform has progressed beyond laboratory testing and being used in offshore wind farm planning, floating wind farm design and engineering, and environmental impact assessments focusing on turbidity. These also will provide feedback for iterative refinement and validation.

1) Research Directions: Advances in AI and machine learning should be leveraged to improve the predictive capabilities of these systems, enabling more accurate scenario modeling for planning decisions that consider the full three-dimensional complexity of maritime environments.

The validation of procedurally generated maritime environments against real-world conditions requires continued attention. Developing robust metrics for assessing the accuracy and ecological validity of generated environments will be essential for building trust in these systems among stakeholders and decision-makers who are accustomed to traditional 2D representations.

VI. Conclusion

This paper presented Procedural Ocean, a novel platform that integrates procedural content generation with serious gaming principles to address fundamental limitations in environmental decision-making tools. Through the development and evaluation of this system, we demonstrated how procedural techniques can efficiently transform geospatial datasets into interactive 3D environments while maintaining both data fidelity and real-time performance, enabling stakeholder engagement that transcends traditional 2D GIS limitations.

Our architecture comprising of data management, procedural generation, and user interaction layers, provides a scalable technical framework for serious games development in complex domains. The platform successfully integrates standard geospatial formats with real-time procedural generation, while the modular Procedural Rule Pack system enables extensible behavior modeling and regional customization. The primary technical innovation lies in bridging the gap between scientific data accuracy and interactive serious games. By developing architectural patterns that combine blackboard data management with procedural generation pipelines, we created a reusable framework for serious games using real-world data. The modular design enables community-driven development and adaptation to diverse environments.

However, our work also revealed ongoing challenges in data standardization, simulation and behavioral model validation, and collaborative functionality. These limitations point to important areas for future technical development, including multi-user architecture design, enhanced uncertainty modeling, and standardized protocols for procedural behavior generation in serious gaming contexts.

Procedural Ocean demonstrates that advanced procedural generation techniques can serve serious environmental gaming applications, providing a technical foundation for similar tools across domains requiring stakeholder engagement with complex spatial data. By establishing architectural patterns for integrating scientific datasets with interactive gaming environments, this work contributes to the broader serious games

development community and opens pathways for procedural approaches in other domains.

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